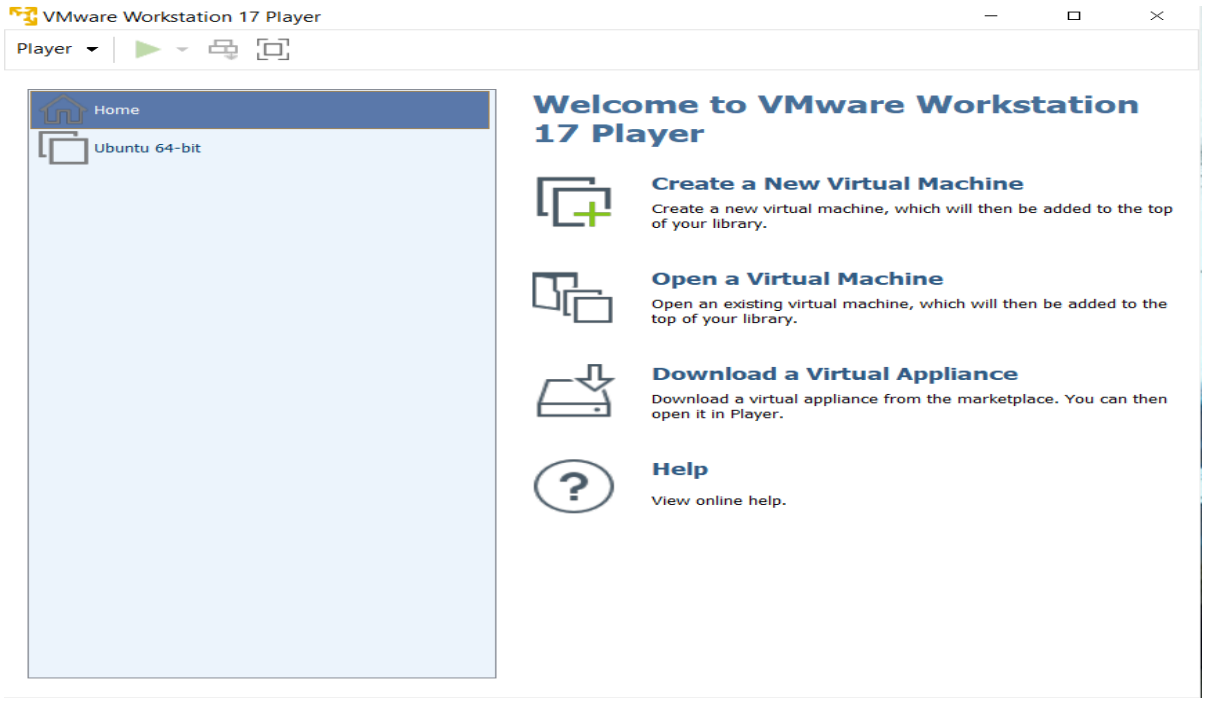
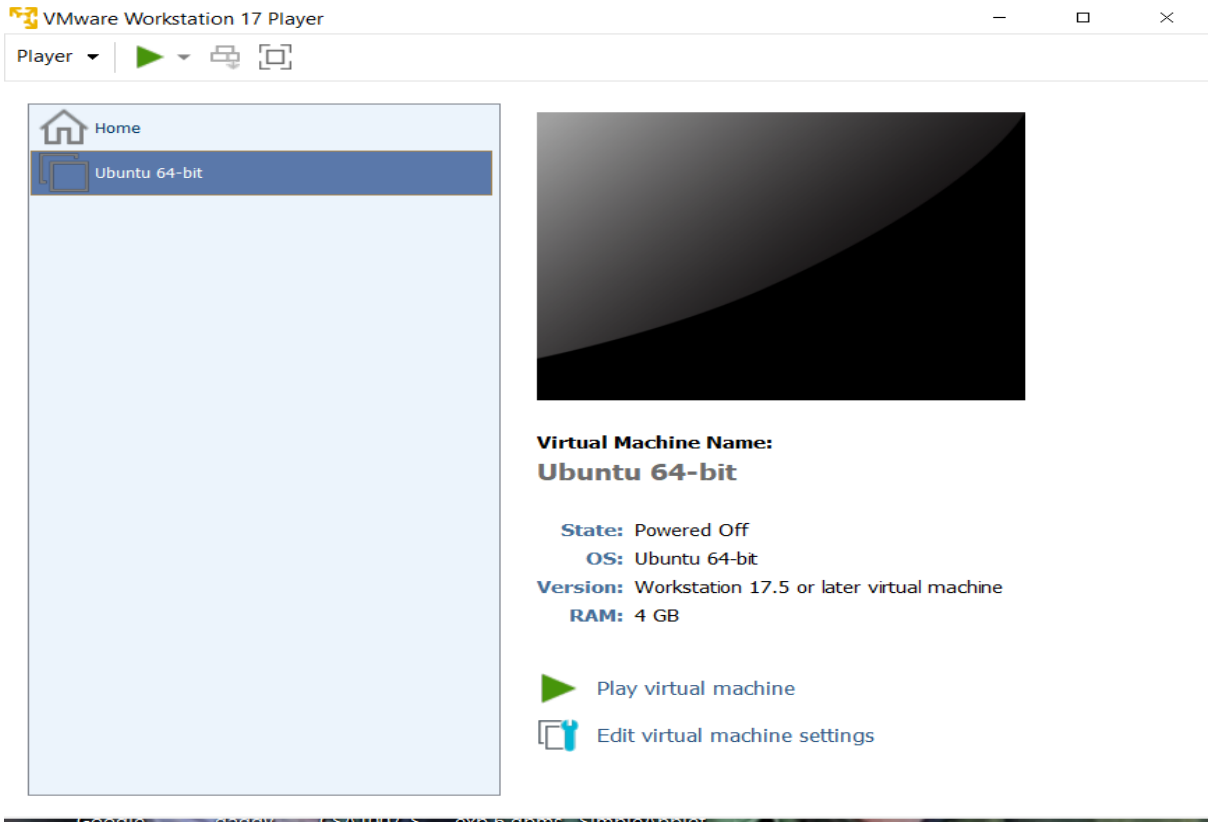
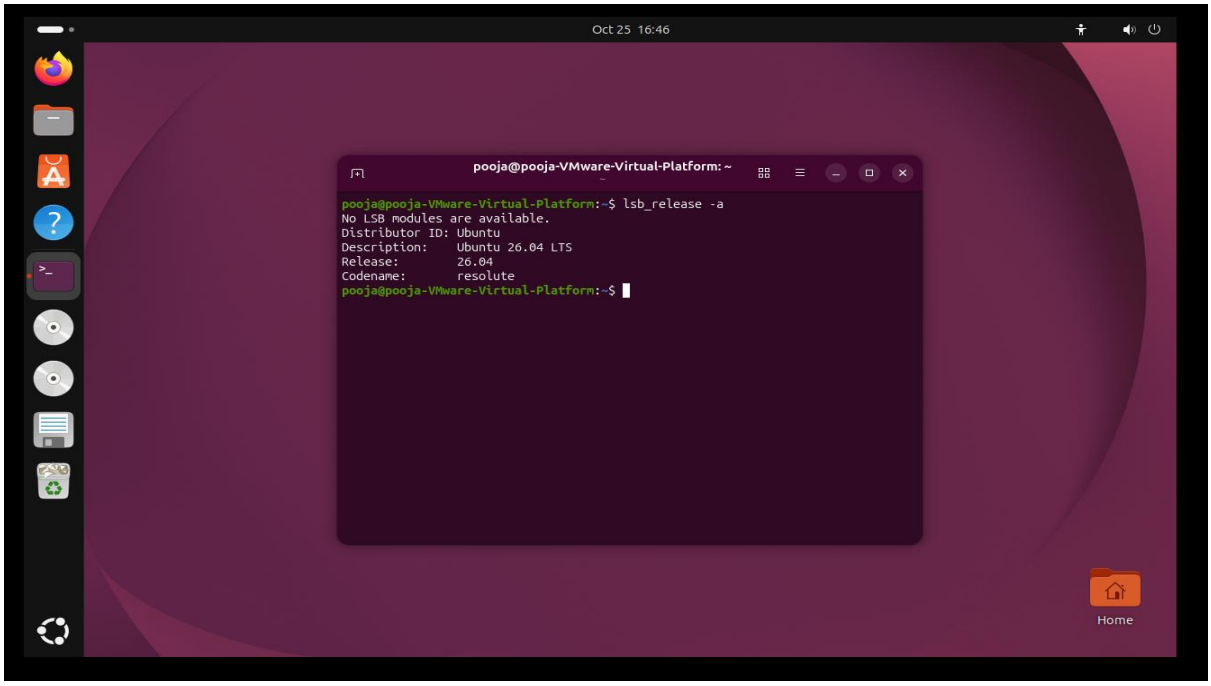
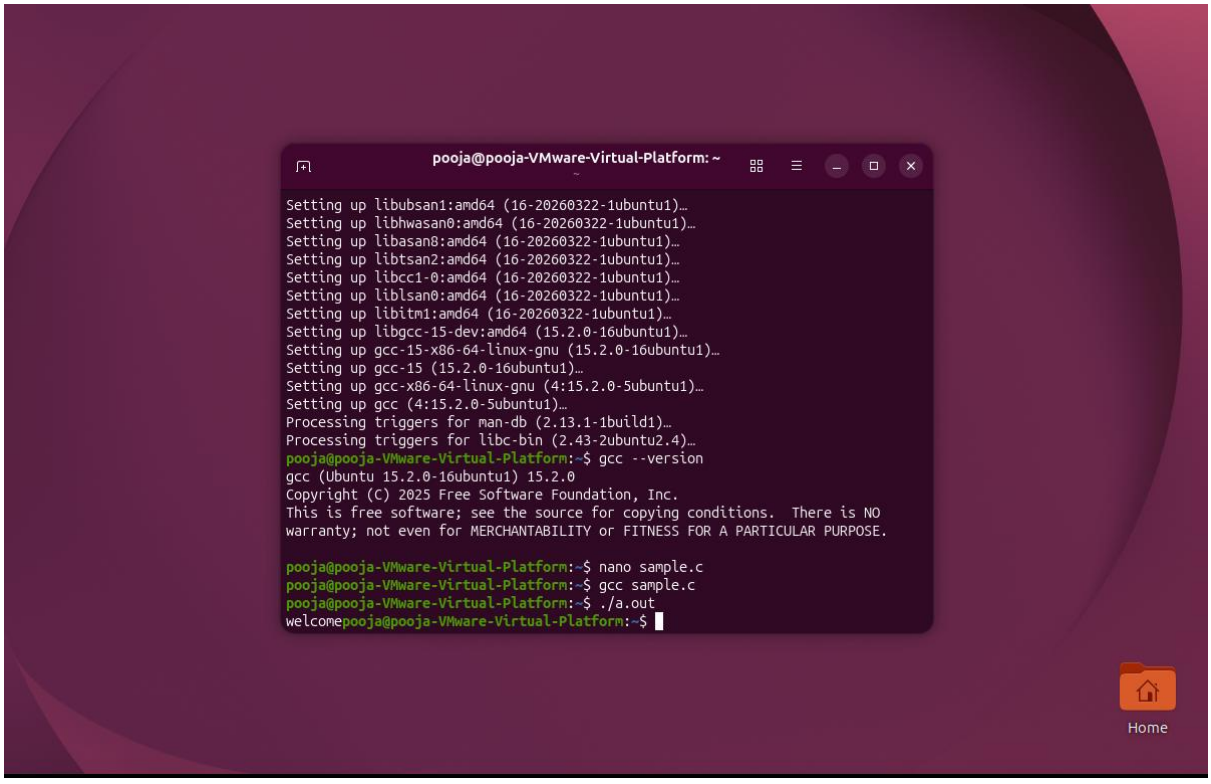
Experiment 18

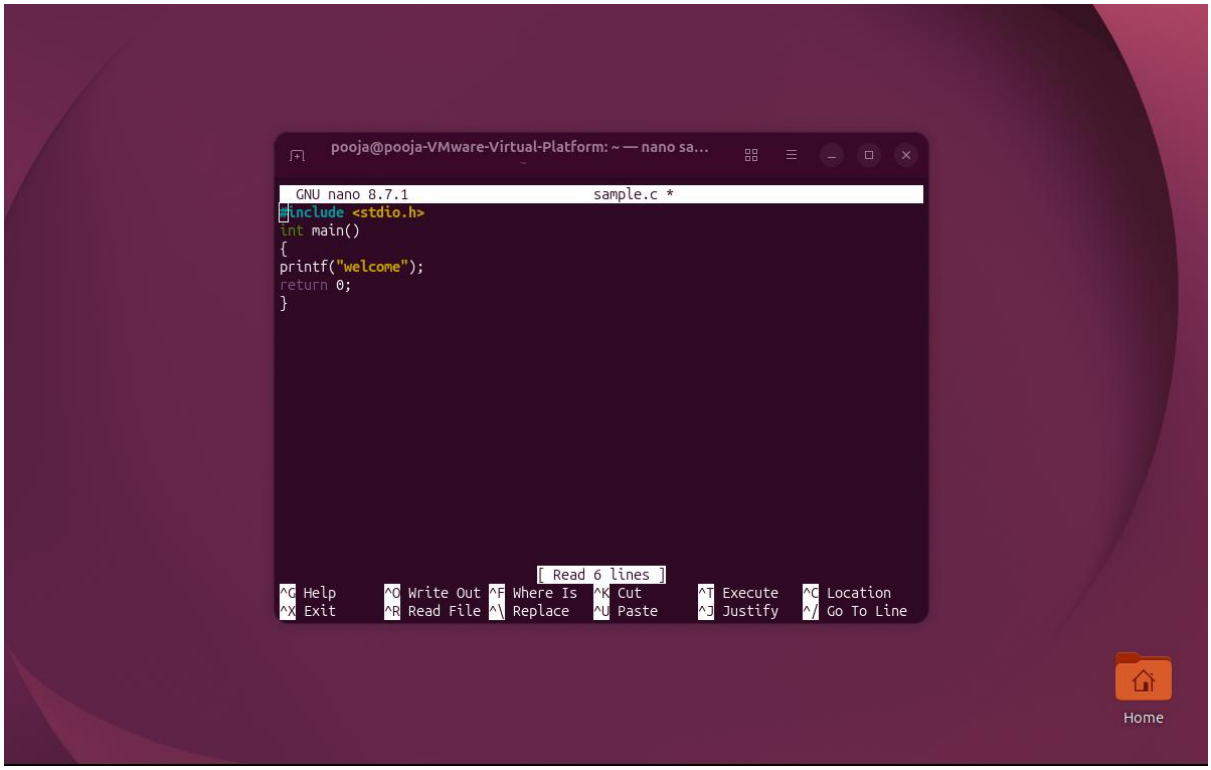


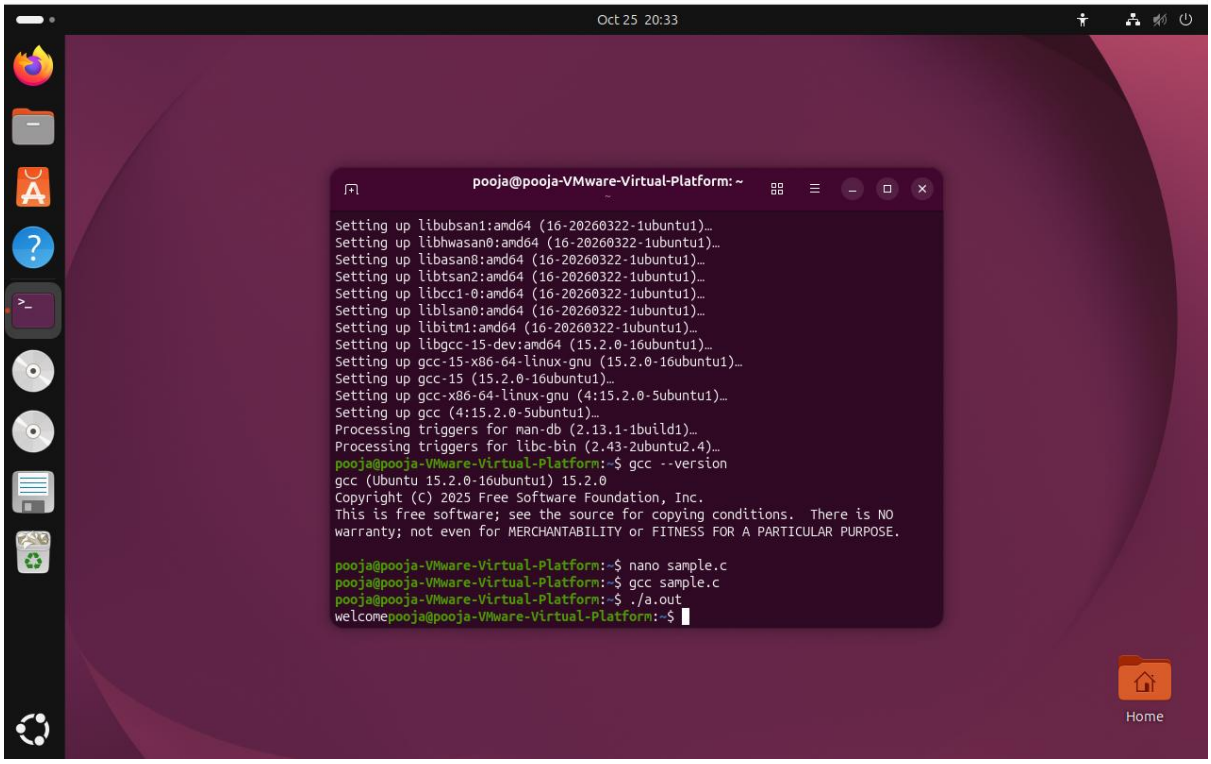




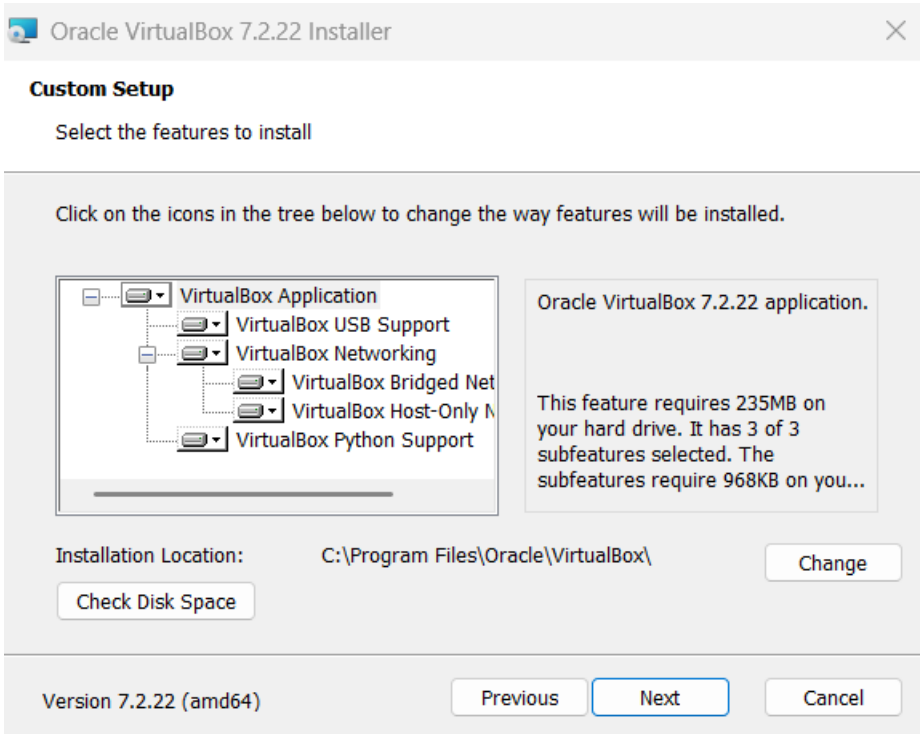
Experiment 19

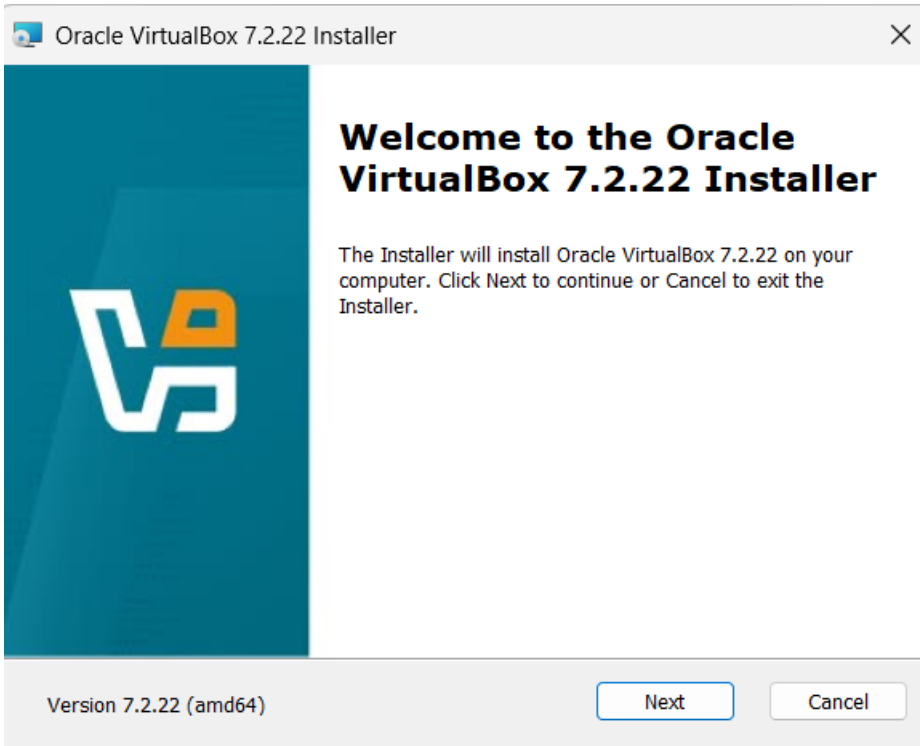



pooja@pooja-VMware-Virtual-Platform: ~ — nano sa...
GNU nano 8.7.1 sample.c *
#include <stdio.h>
int main()
{
printf("welcome");
return 0;
}
[Read 6 lines]
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line
Home

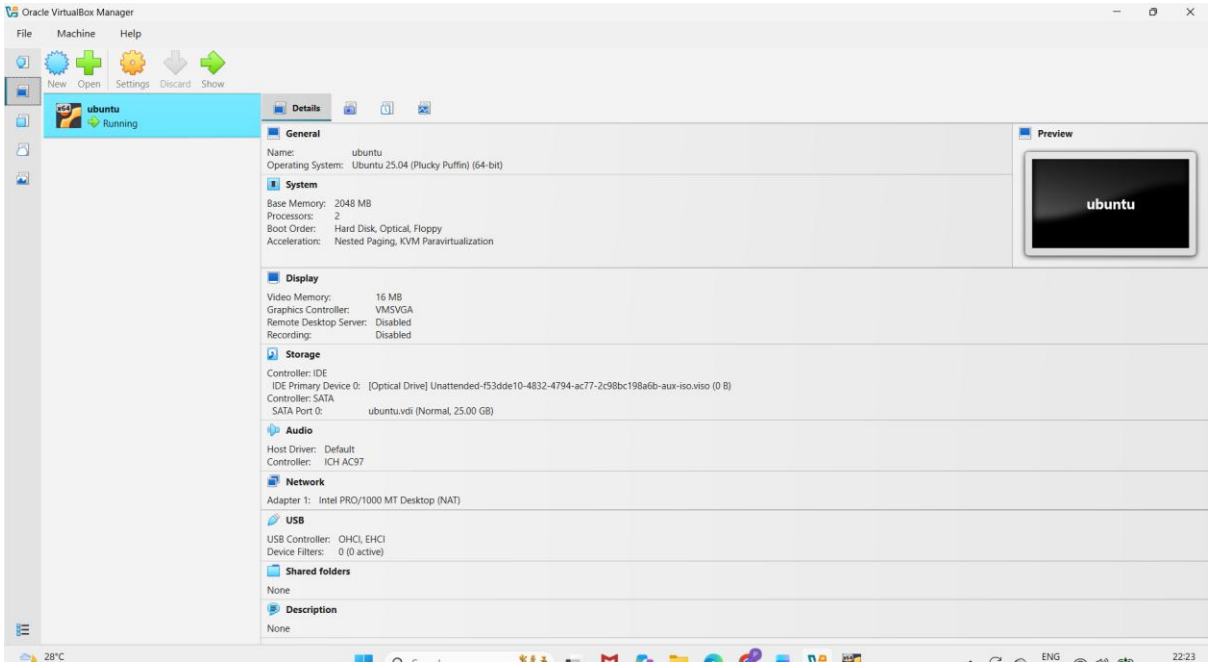

Oct 25 20:33
pooja@pooja-VMware-Virtual-Platform: ~
Setting up libubsan1:amd64 (16-20260322-1ubuntu1)…
Setting up libhwasan0:amd64 (16-20260322-1ubuntu1)…
Setting up libasan8:amd64 (16-20260322-1ubuntu1)…
Setting up libtsan2:amd64 (16-20260322-1ubuntu1)…
Setting up libcc1-0:amd64 (16-20260322-1ubuntu1)…
Setting up liblsan0:amd64 (16-20260322-1ubuntu1)…
Setting up libitm1:amd64 (16-20260322-1ubuntu1)…
Setting up libgcc-15-dev:amd64 (15.2.0-16ubuntu1)…
Setting up gcc-15-x86-64-linux-gnu (15.2.0-16ubuntu1)…
Setting up gcc-15 (15.2.0-16ubuntu1)…
Setting up gcc-x86-64-linux-gnu (4:15.2.0-5ubuntu1)…
Setting up gcc (4:15.2.0-5ubuntu1)…
Processing triggers for man-db (2.13.1-1build1)…
Processing triggers for libc-bin (2.43-2ubuntu2.4)…
pooja@pooja-VMware-Virtual-Platform:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
pooja@pooja-VMware-Virtual-Platform:~$ nano sample.c
pooja@pooja-VMware-Virtual-Platform:~$ gcc sample.c
pooja@pooja-VMware-Virtual-Platform:~$ ./a.out
welcomepooja@pooja-VMware-Virtual-Platform:~$
Home

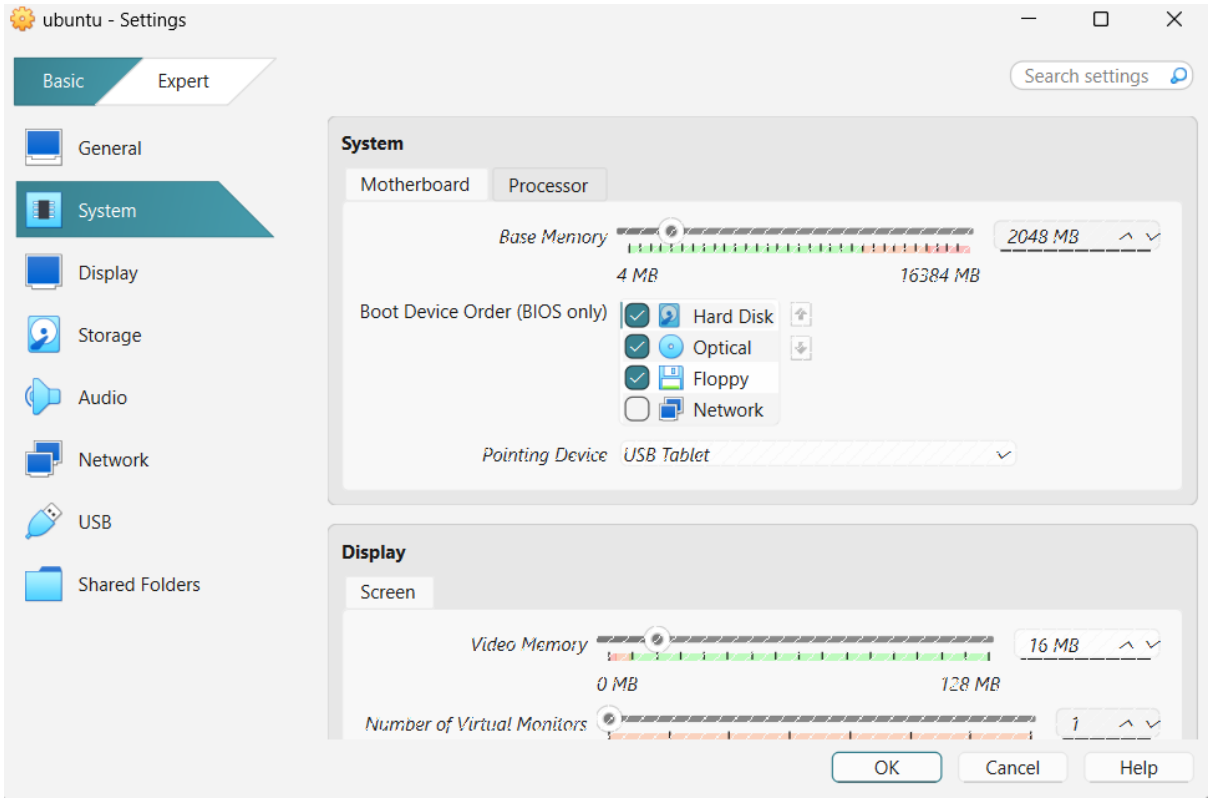
Experiment 20 & 21

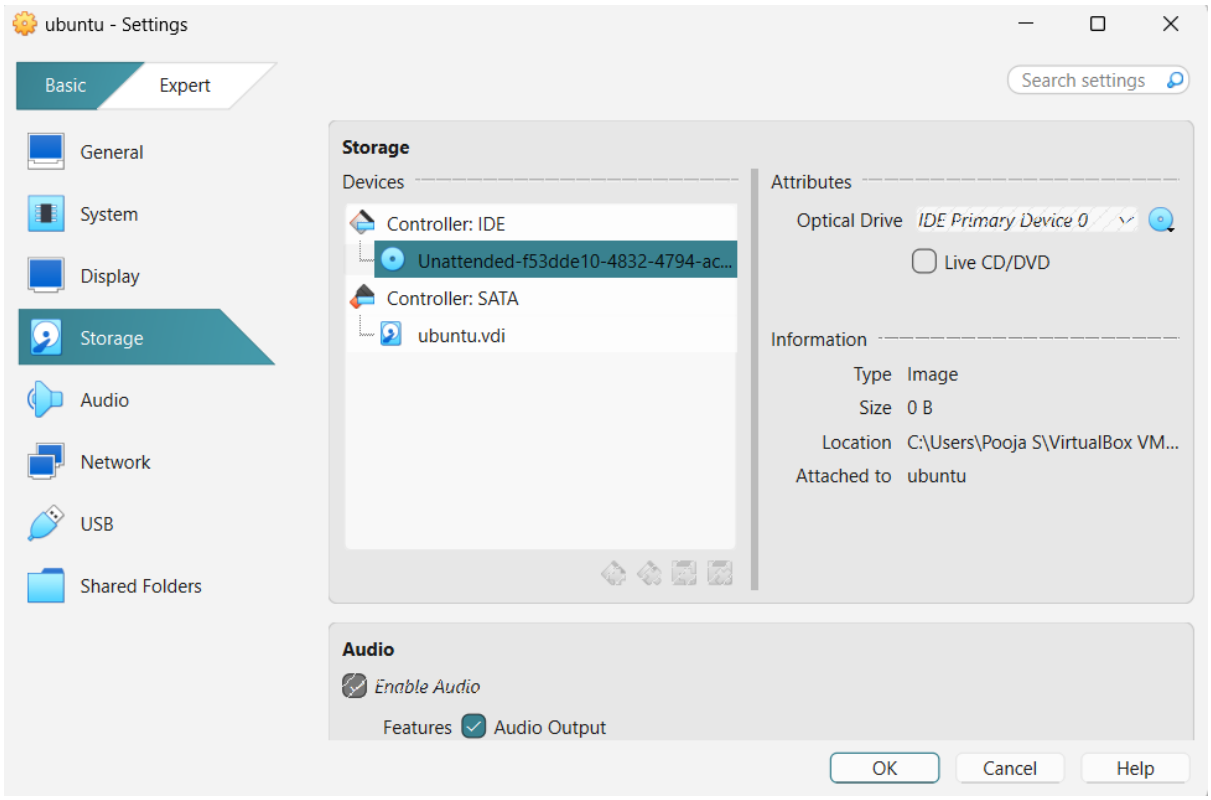


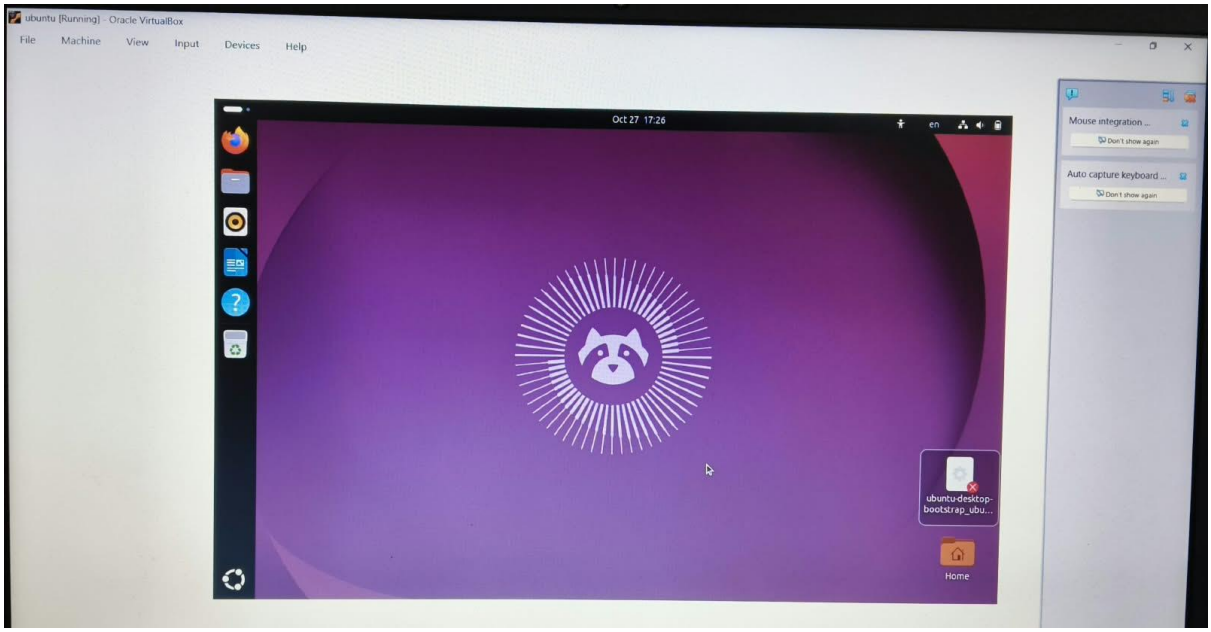



Oracle VirtualBox 7.1.8 Setup
Oracle VirtualBox 7.1.8 installation is complete.
Click the Finish button to exit the Setup Wizard.
Version 7.1.8
< Back
Finish
Cancel

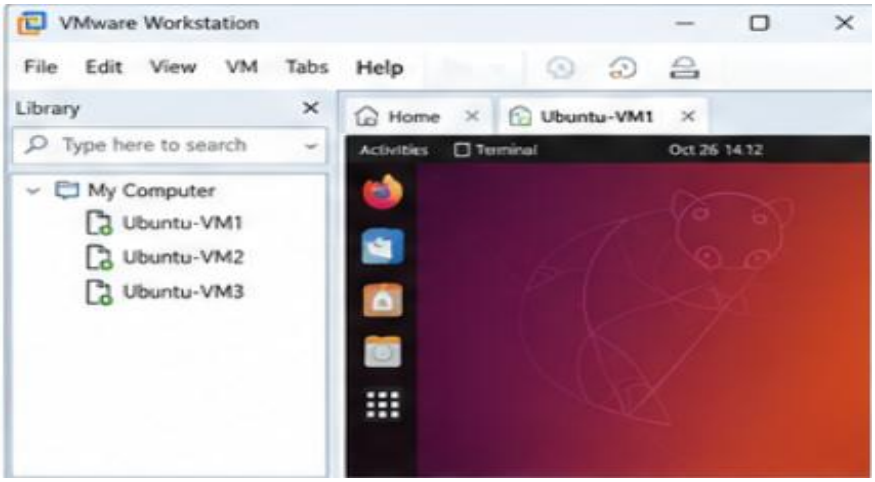

Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings
Discard
Show
ubuntu
Running
Details
General
Name: ubuntu
Operating System: Ubuntu 25.04 (Plucky Puffin) (64-bit)
System
Base Memory: 2048 MB
Processors: 2
Boot Order: Hard Disk, Optical, Floppy
Acceleration: Nested Paging, KVM Paravirtualization
Preview
ubuntu
Display
Video Memory: 16 MB
Graphics Controller: VMSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: IDE
IDE Primary Device 0: [Optical Drive] Unattended-f53dde10-4832-4794-ac77-2c98bc198a6b-aux-iso.viso (0 B)
Controller: SATA
SATA Port 0: ubuntu.vdi (Normal, 25.00 GB)
Audio
Host Driver: Default
Controller: ICH AC97
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)
USB
USB Controller: OHCI, EHCI
Device Filters: 0 (0 active)
Shared folders
None
Description
None
28°C
ENG
22:23

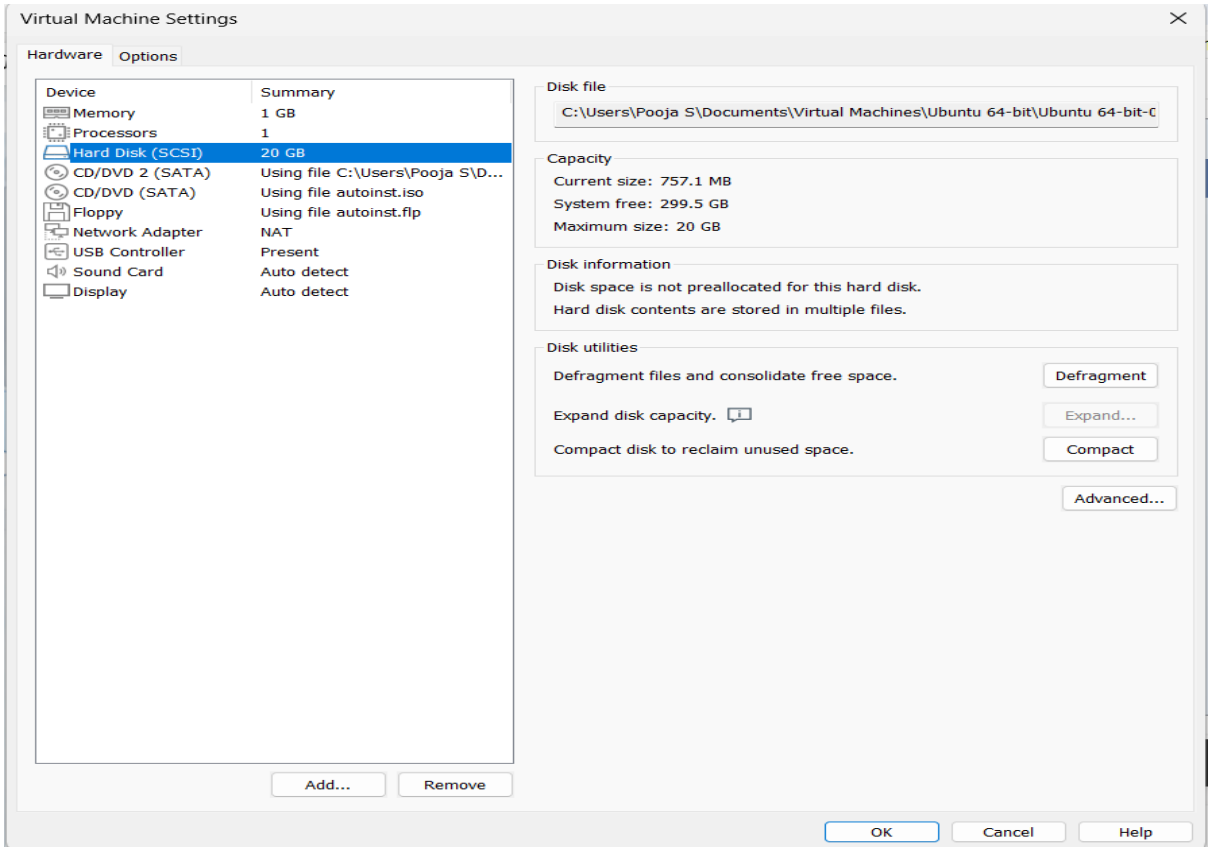

ubuntu - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
USB
Shared Folders
System
Motherboard
Processor
Base Memory
2048 MB
4 MB
16384 MB
Boot Device Order (BIOS only)
Hard Disk
Optical
Floppy
Network
Pointing Device
USB Tablet
Display
Screen
Video Memory
16 MB
0 MB
128 MB
Number of Virtual Monitors
1
OK
Cancel
Help

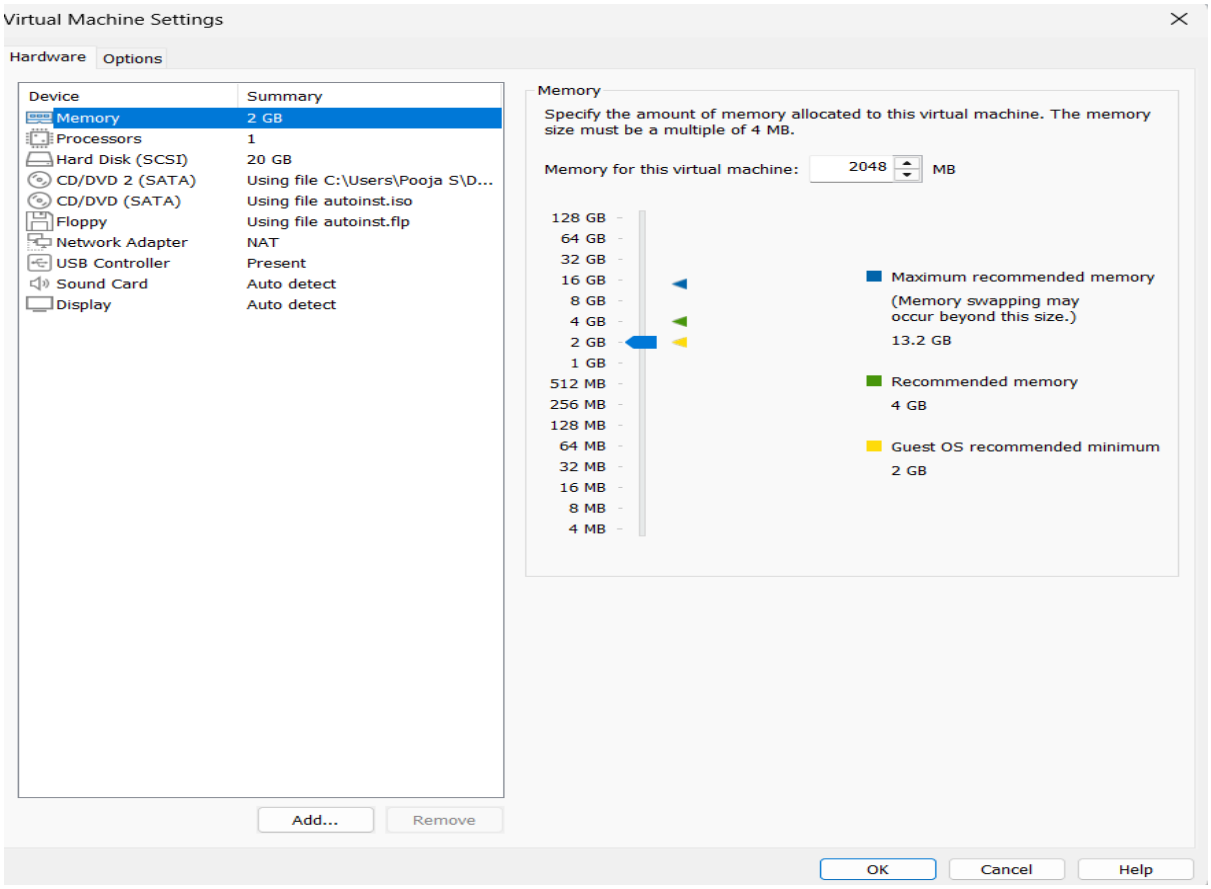

ubuntu - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
USB
Shared Folders
Storage
Devices
Controller: IDE
Unattended-f53dde10-4832-4794-ac...
Controller: SATA
ubuntu.vdi
Attributes
Optical Drive
IDE Primary Device 0
Live CD/DVD
Information
Type Image
Size 0 B
Location C:\Users\Pooja S\VirtualBox VM...
Attached to ubuntu
Audio
Enable Audio
Features
Audio Output
OK
Cancel
Help

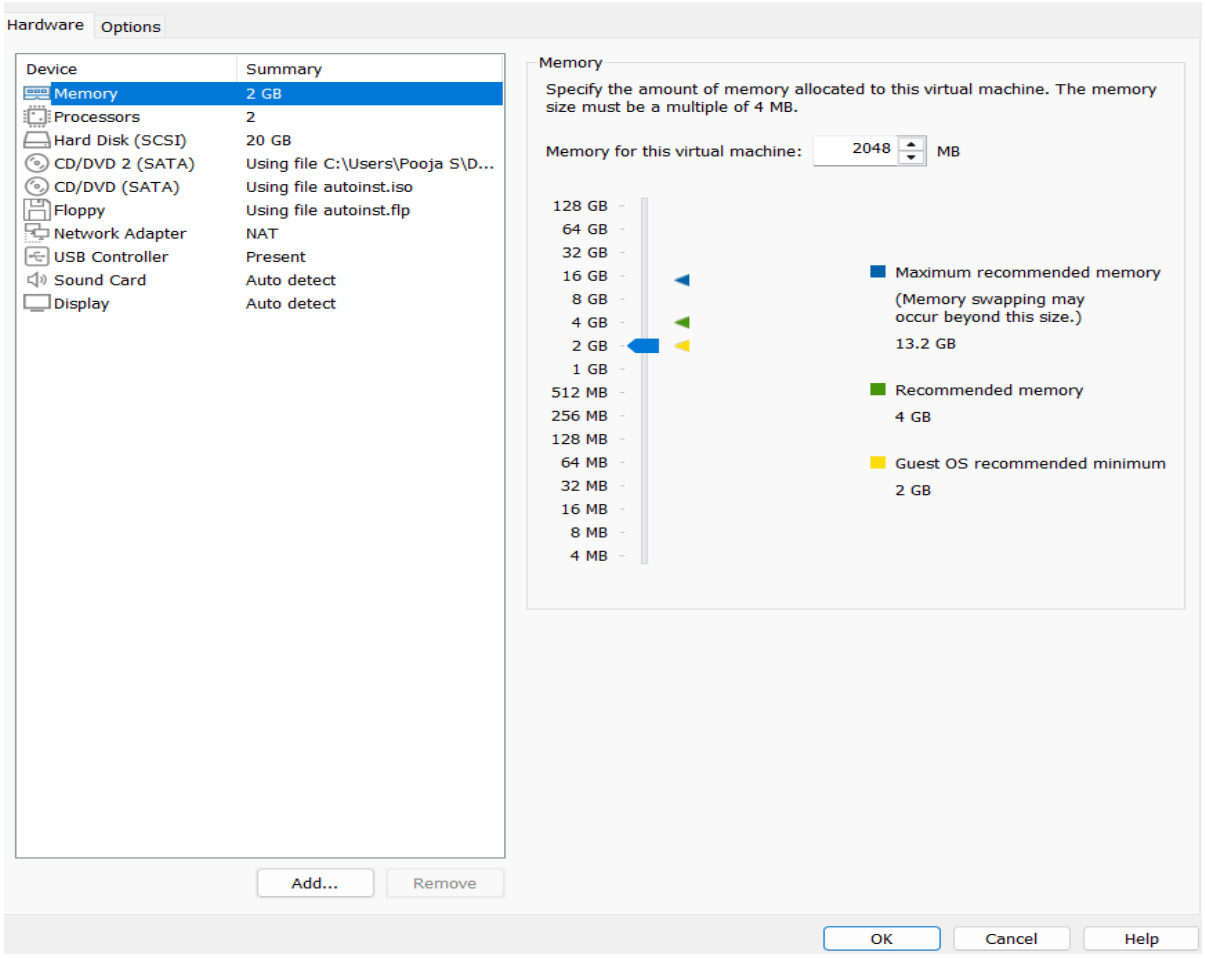


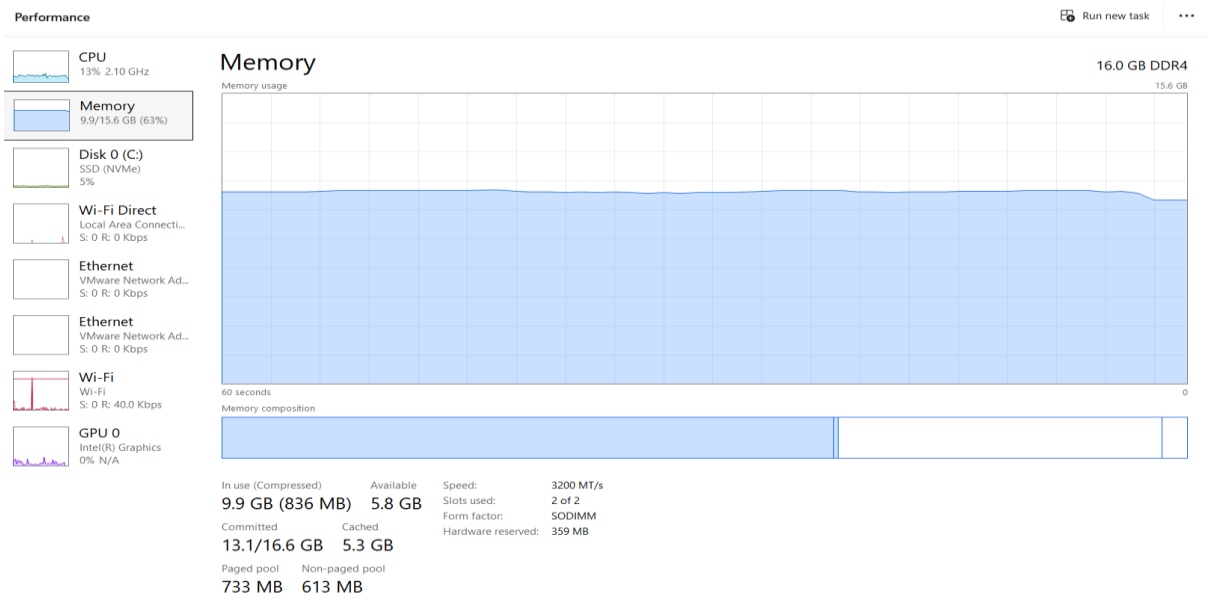
Experiment -22

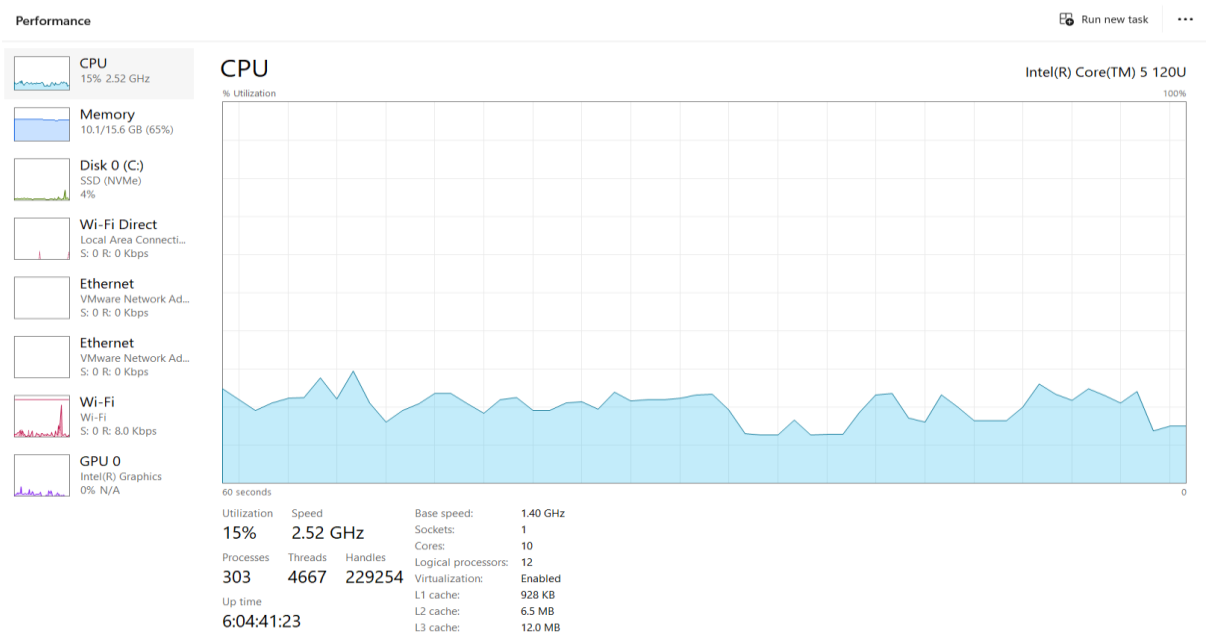


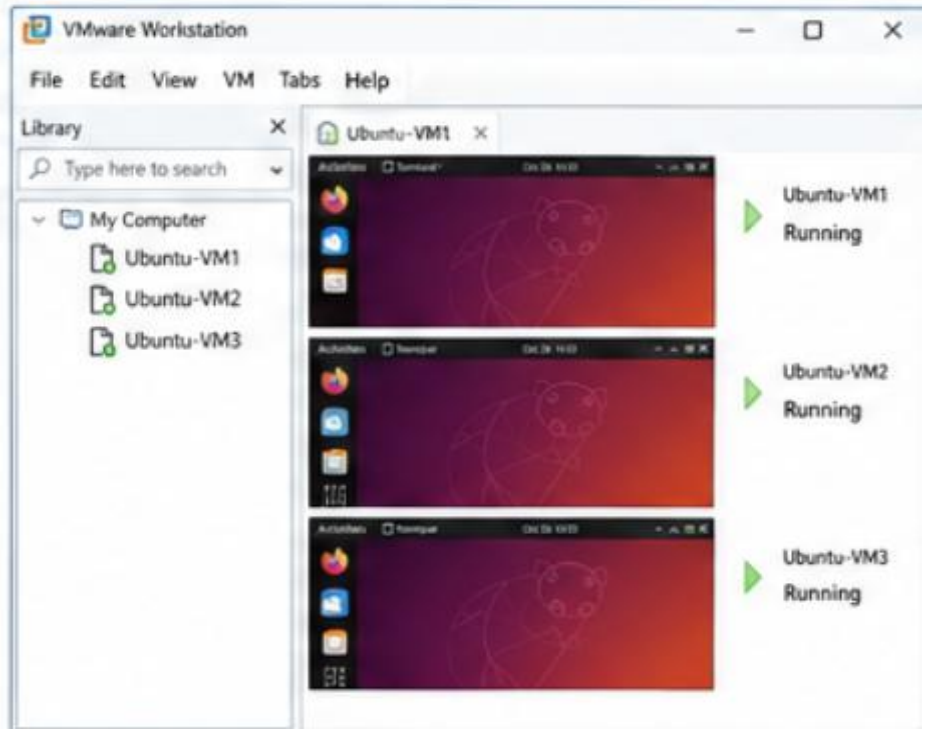

Virtual Machine Settings
Hardware
Options
Device
Summary
Memory
1 GB
Processors
1
Hard Disk (SCSI)
20 GB
CD/DVD 2 (SATA)
Using file C:\Users\Pooja S\D...
CD/DVD (SATA)
Using file autoinst.iso
Floppy
Using file autoinst.flp
Network Adapter
NAT
USB Controller
Present
Sound Card
Auto detect
Display
Auto detect
Add...
Remove
Disk file
C:\Users\Pooja S\Documents\Virtual Machines\Ubuntu 64-bit\Ubuntu 64-bit-(
Capacity
Current size: 757.1 MB
System free: 299.5 GB
Maximum size: 20 GB
Disk information
Disk space is not preallocated for this hard disk.
Hard disk contents are stored in multiple files.
Disk utilities
Defragment files and consolidate free space.
Defragment
Expand disk capacity.
Expand...
Compact disk to reclaim unused space.
Compact
Advanced...
OK
Cancel
Help


Virtual Machine Settings
Hardware
Options
Device
Summary
Memory
2 GB
Processors
1
Hard Disk (SCSI)
20 GB
CD/DVD 2 (SATA)
Using file C:\Users\Pooja S\D...
CD/DVD (SATA)
Using file autoinst.iso
Floppy
Using file autoinst.flp
Network Adapter
NAT
USB Controller
Present
Sound Card
Auto detect
Display
Auto detect
Add...
Remove
Memory
Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.
Memory for this virtual machine:
2048
MB
128 GB
64 GB
32 GB
16 GB
8 GB
4 GB
2 GB
1 GB
512 MB
256 MB
128 MB
64 MB
32 MB
16 MB
8 MB
4 MB
Maximum recommended memory
(Memory swapping may occur beyond this size.)
13.2 GB
Recommended memory
4 GB
Guest OS recommended minimum
2 GB
OK
Cancel
Help


Hardware
Options
Device
Summary
Memory
2 GB
Processors
2
Hard Disk (SCSI)
20 GB
CD/DVD 2 (SATA)
Using file C:\Users\Pooja S\D...
CD/DVD (SATA)
Using file autoinst.iso
Floppy
Using file autoinst.flp
Network Adapter
NAT
USB Controller
Present
Sound Card
Auto detect
Display
Auto detect
Add...
Remove
Memory
Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.
Memory for this virtual machine:
2048
MB
128 GB
64 GB
32 GB
16 GB
8 GB
4 GB
2 GB
1 GB
512 MB
256 MB
128 MB
64 MB
32 MB
16 MB
8 MB
4 MB
Maximum recommended memory
(Memory swapping may occur beyond this size.)
13.2 GB
Recommended memory
4 GB
Guest OS recommended minimum
2 GB
OK
Cancel
Help


Performance
Run new task
CPU
13% 2.10 GHz
Memory
9.9/15.6 GB (63%)
Disk 0 (C:)
SSD (NVMe)
5%
Wi-Fi Direct
Local Area Connecti...
S: 0 R: 0 Kbps
Ethernet
VMware Network Ad...
S: 0 R: 0 Kbps
Ethernet
VMware Network Ad...
S: 0 R: 0 Kbps
Wi-Fi
Wi-Fi
S: 0 R: 40.0 Kbps
GPU 0
Intel(R) Graphics
0% N/A
Memory
16.0 GB DDR4
Memory usage
15.6 GB
60 seconds
0
Memory composition
In use (Compressed)
9.9 GB (836 MB)
Available
5.8 GB
Committed
13.1/16.6 GB
Cached
5.3 GB
Paged pool
733 MB
Non-paged pool
613 MB
Speed:
3200 MT/s
Slots used:
2 of 2
Form factor:
SODIMM
Hardware reserved:
359 MB


Performance
Run new task
CPU
15% 2.52 GHz
Memory
10.1/15.6 GB (65%)
Disk 0 (C:)
SSD (NVMe)
4%
Wi-Fi Direct
Local Area Connecti...
S: 0 R: 0 Kbps
Ethernet
VMware Network Ad...
S: 0 R: 0 Kbps
Ethernet
VMware Network Ad...
S: 0 R: 0 Kbps
Wi-Fi
Wi-Fi
S: 0 R: 8.0 Kbps
GPU 0
Intel(R) Graphics
0% N/A
CPU
Intel(R) Core(TM) 5 120U
% Utilization
100%
60 seconds
0
Utilization
15%
Speed
2.52 GHz
Processes
303
Threads
4667
Handles
229254
Up time
6:04:41:23
Base speed: 1.40 GHz
Sockets: 1
Cores: 10
Logical processors: 12
Virtualization: Enabled
L1 cache: 928 KB
L2 cache: 6.5 MB
L3 cache: 12.0 MB

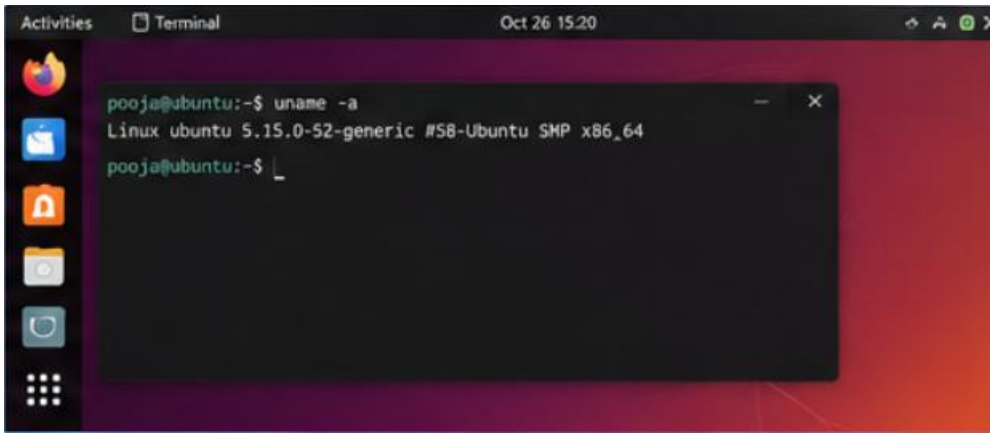

VMware Workstation
File
Edit
View
VM
Tabs
Help
Library
Type here to search
My Computer
Ubuntu-VM1
Ubuntu-VM2
Ubuntu-VM3
Ubuntu-VM1
Ubuntu-VM1
Running
Ubuntu-VM2
Ubuntu-VM2
Running
Ubuntu-VM3
Ubuntu-VM3
Running

VM Configuration and Observation

VM	RAM	CPU Cores	Storage	Status
VM1	1 GB	1	10 GB	Running
VM2	2 GB	1	15 GB	Running
VM3	2 GB	2	20 GB	Running
VM4	2 GB	2		May become slow / insufficient resources

System Resources (example)

Resource	Utilization	Available
CPU	70 – 85%	15 – 30%
Memory	60 – 75%	20 – 35%
Disk	30 – 50%	–



Result

Virtual machines with different configurations (RAM, CPU and storage) were successfully created and executed. The number of VMs that can run simultaneously depends mainly on the available RAM and CPU resources of the host machine.